Chapter 7

Circles, Part 2 Of 2

In this chapter we conclude the study of how inversion of the real plane acts on straight lines and circles by covering the remaining cases when a preimage circle does not pass through the center of inversion and is positioned with respect to the circle of inversion in such a way that the two circles at hand either touch each other, internally or externally, or do not have any points in common at all.

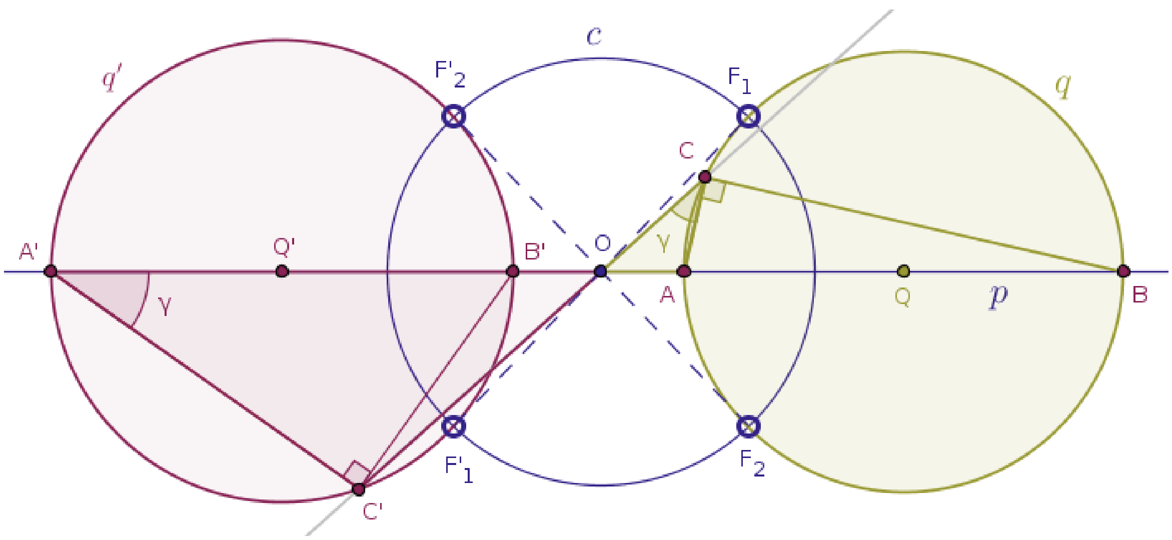
In the previous chapter we showed a complete proof of the Theorem 23. That proof can be abstracted into the following template:

- draw the straight line of centers, p
- invert the points, A and B, at which the above line of centers, p, intersects the given circle, q, under a need power, positive or negative, in a given circle, c, into their images, A' and B'
- invert a random point, C, on the given circle, q, with a needed power into its image, C'
- connect that image point, C', with the image points A' and B'
- use the Theorem 22 (ICSTT) in order to show that the triangles OAC and $OC'A'$ are similar
- use the fact that the angle $OB'C'$ is exterior for the triangle $B'C'A'$
- use the fact that the angles OCB is composite: $\angle OCB = \gamma + 90^\circ$

- show that the angle $B'C'A'$ must be right because $\angle B'C'A' + \gamma$ must be equal to $\gamma + 90^\circ$ since the magnitude of an exterior angle of a triangle is equal to the sum of the magnitudes of the opposite and interior angles of that triangle

We urge our readers to use the above template in order to generate the proofs of the following six related theorems on their own. Meanwhile, we will only formulate these theorems and highlight one or two relevant triangles and angles in the pictorial demonstrations of these theorems.

Theorem 24. Under the inversion of a plane with respect to a circle, $c(O, r)$, with negative power a circle, $q(Q, R)$, that does not pass through the center of inversion, O, and has exactly two distinct points, F_1 and F_2, common with the circle of inversion, c, is transformed into a circle, $q'(Q', R')$, that does not pass through the center of inversion, O, but does pass through the points, F_1' and F_2', which are the diametrically opposite images of the common points, F_1 and F_2, correspondingly (Figure 7.1):



Proof. Exercise (use the template described above in order to put a proof of this theorem together on your own). □

Observe that even in this case, when the power of inversion is negative, our template described above still works and, amazingly enough, pretty much verbatim.

From the above theorem it immediately follows that in order to construct the image of a circle $q(Q, R)$ under the inversion of a plane with respect to a circle $c(O, r)$ with negative power when the circle q does not pass through the center of inversion, O, and has exactly two distinct points, F_1 and F_2, common with the circle of inversion, c, with Euclidean tools alone, we carry out the following steps.

Draw a straight line, p, through the center of inversion, O, and the center, Q, of the given circle, q.

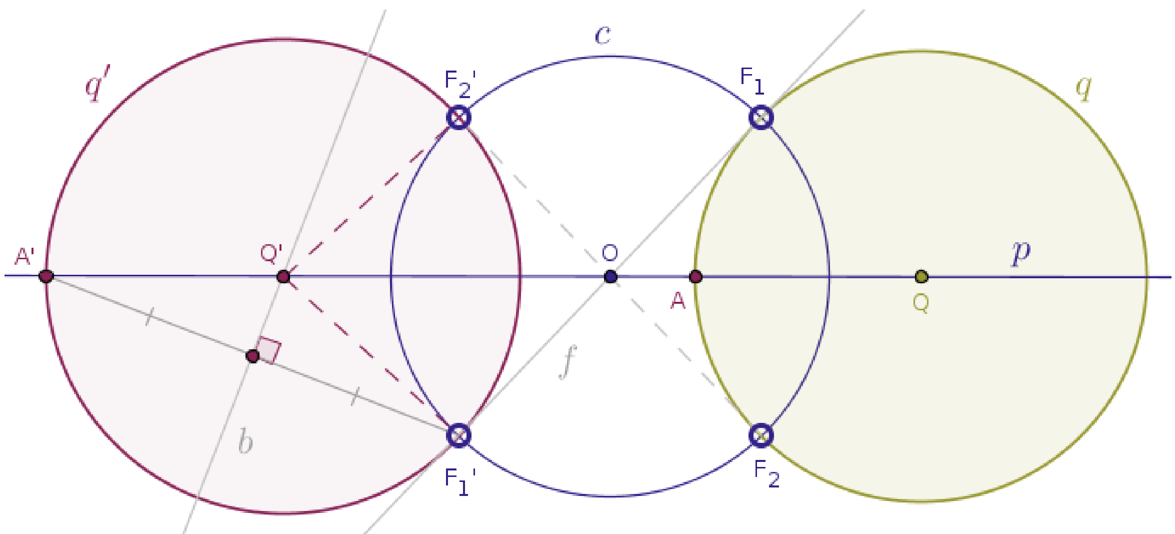
That straight line (of centers), p, will intersect the given circle, q, at two distinct and diametrically opposite points, of which we need only one. Name that one point as A.

Reflect the point A in the circle of inversion, c, with negative power into its image point, A'.

Draw a straight line, f, through the center of inversion, O, and one the points, F_1 (or F_2) , that is common for the circle of inversion, c, and the given circle, q.

That straight line, f, will intersect the circle of inversion, c, for the second time at a point, F_1' (or F_2'), that is diametrically opposite to the chosen common point, F_1 (or F_2).

Bisect the line segment $A'F_1'$ (or $A'F_2'$) with a straight line, b, until that straight line intersects the line of centers, p, at a point, Q', that is the center of the sought-after image circle, q' (Figure 7.2):



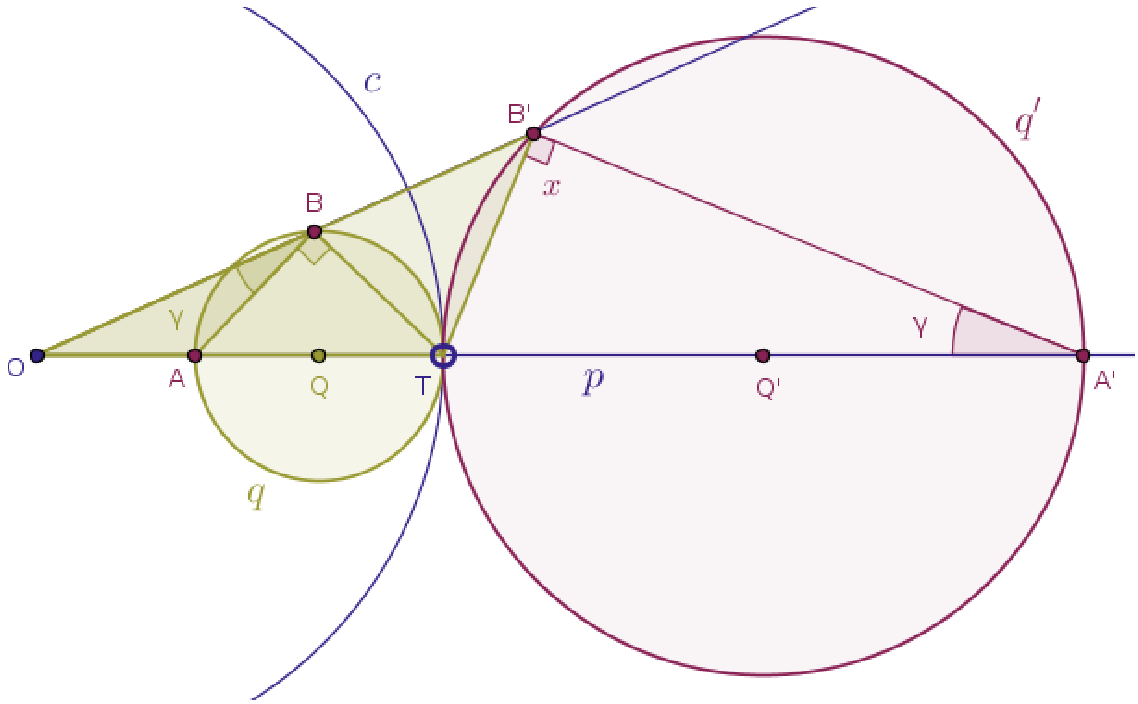
Draw the circle, q', centered at the point said point Q' with the radius $|Q'A'| = |Q'F_1'| = |Q'F_2'|$.

The circle $q'(Q', |Q'A'| = |Q'F_1'| = |Q'F_2'|)$ thus constructed is the image of the circle q.

Proof. Exercise. □

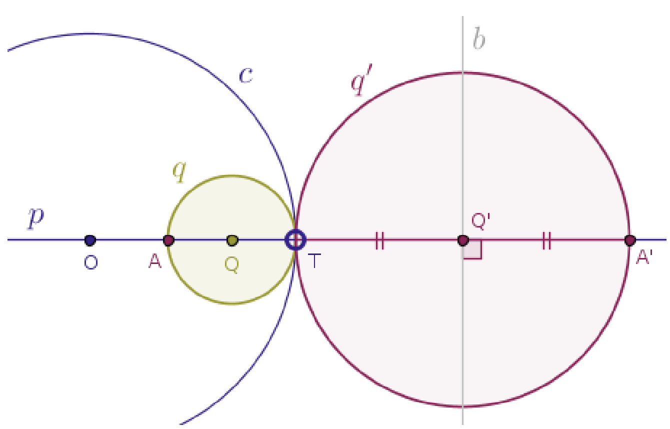
Here, also use the said template but argue from the fact that q' **is** a circle ...

Theorem 25. Under the inversion of a plane with respect to a circle, $c(O, r)$, with positive power a (green) circle, $q(Q, R)$, that does not pass through the center of inversion, O, and touches the circle of inversion, c, at a point, T, internally is transformed into a (red) circle, $q'(Q', R')$ that touches the circle of inversion, c, at the point of tangency, T, externally. (Figure 7.3):



Proof. Exercise (here note that a random point C on a given circle q became B because the role of the point B in our template is played by the tangency point, T. Otherwise, by ICSTT, the triangles OAB and $OB'AB'$ are similar and so are the triangles OBT and OTB' and so on). □

From the above theorem it immediately follows that in order to construct the image of a circle $q(Q, R)$ under the inversion of a plane with respect to a circle $c(O, r)$ with positive power when the circle q does not pass through the center of inversion, O, and touches the circle of inversion, c, at a point, T, internally with Euclidean tools alone, we carry out the following steps (Figure 7.4):



Draw a straight line, p, through the center of inversion, O, and the center, Q, of the given circle, q.

That straight line (of centers), p, will intersect the given circle, q, at two distinct and diametrically opposite points, of which one will be the point of tangency, T, and the other will be a point that lies closer to the center of inversion, O, than the tangency point, T. We need that, second, point. Name that point as A.

Reflect the point A in the circle of inversion, c, with positive power into its image point, A'.

Bisect the line segment TA' with a straight line, b, until that straight line intersects the line of centers, p, at a point, Q', that is the center of the sought-after image circle, q'.

Draw the circle, q', centered at the point said point Q' with the radius $|Q'A'| = |Q'T|$.

The circle $q'(Q', |Q'A'| = |Q'T|)$ thus constructed is the image of the circle q.

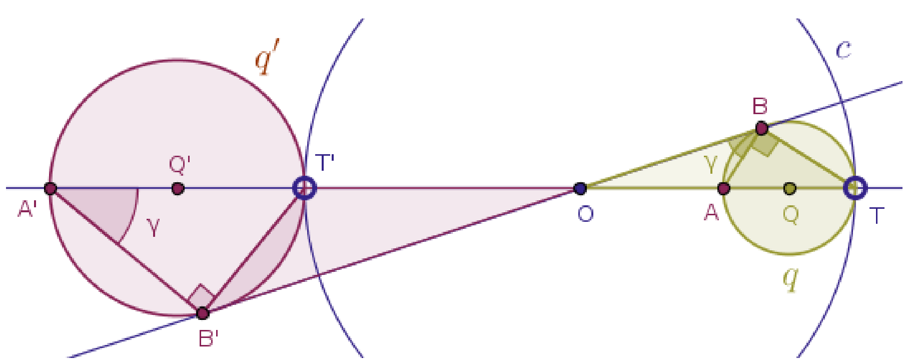
Proof. Exercise. □

The following theorem is the converse of the Theorem 25.

Theorem 26. Under the inversion of a plane with respect to a circle, $c(O, r)$, with positive power a (green) circle, $q(Q, R)$, that does not pass through the center of inversion, O, and touches the circle of inversion, c, at a point, T, **ex**ternally is transformed into a (red) circle, $q'(Q', R')$ that touches the circle of inversion, c, at the point of tangency, T, **in**ternally.

Proof. Exercise (note that in this theorem the circles q and q' from the Theorem 25 simply trade places. Thus, for a proof of this theorem rename the circle q into q' and conversely, rename the circle q' into q, rename the point A into A' and conversely, rename the point A' into A). □

Theorem 27. Under the inversion of a plane with negative power (Figure 7.5):



with respect to a circle, $c(O, r)$, a circle, $q(Q, R)$, that does not pass through the center of inversion, O, and touches the circle of inversion, c, at a point, T, internally is transformed into a circle, $q'(Q', R')$, that touches the circle of inversion, c, externally at a point, T', which is a diametrically opposite image of the tangency point, T.

Proof. Exercise. □

From the above theorem it immediately follows that in order to construct the image of a circle $q(Q, R)$ under the inversion of a plane with respect to a circle $c(O, r)$ with negative power when the circle q does not pass through the center of inversion, O, and touches the circle of inversion, c, at a point, T, internally with Euclidean tools alone, we carry out the following steps.

Draw a straight line, p, through the center of inversion, O, and the center, Q, of the given circle, q.

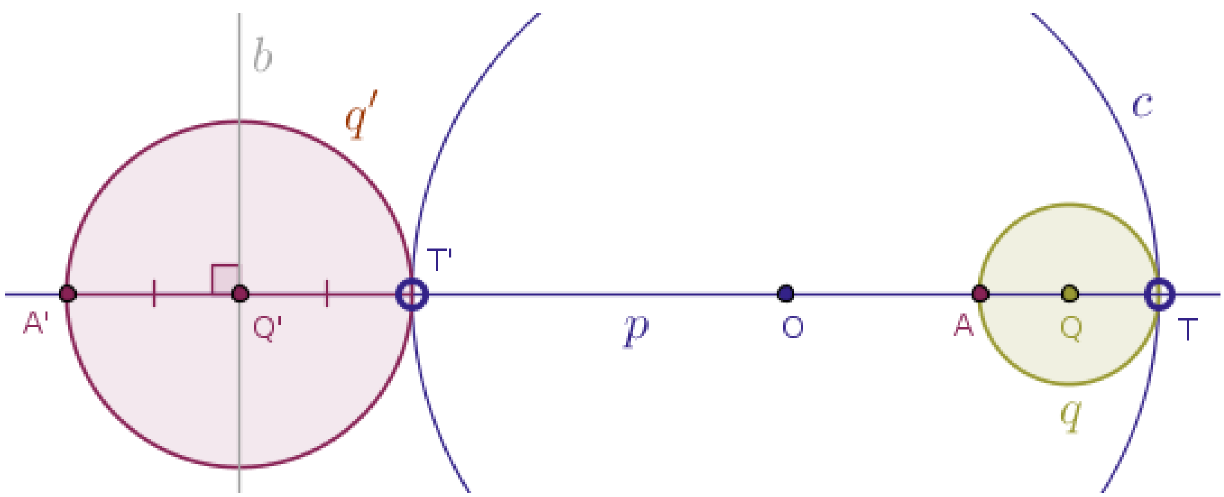
That straight line (of centers), p, will intersect the given circle, q, at two distinct and diametrically opposite points, of which one will be the point of tangency, T, and the other will be a point that lies closer to the center of inversion, O, than the tangency point, T.

We need that, second, point. Name that point as A.

Reflect the point A in the circle of inversion, c, with negative power into its image point, A'.

Bisect the line segment TA' with a straight line, b, until that straight line intersects the line of centers, p, at a point, Q', that is the center of the sought-after image circle, q'.

Draw the circle, q', centered at the point said point Q' with the radius $|Q'A'| = |Q'T|$ (Figure 7.6):



The circle $q'(Q', |Q'A'| = |Q'T|)$ thus constructed is the image of the circle q.

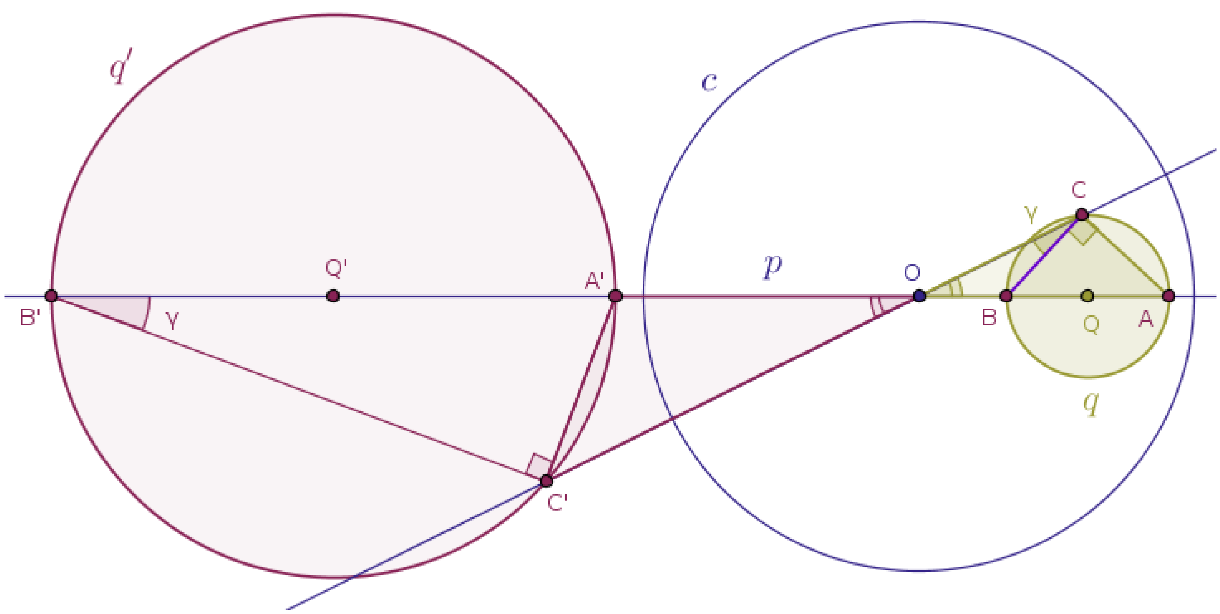
Proof. Exercise. □

The following theorem is the converse of the Theorem 27.

Theorem 28. Under the inversion of a plane with negative power with respect to a circle, $c(O, r)$, a circle, $q(Q, R)$, that does not pass through the center of inversion, O, and touches the circle of inversion, c, at a point, T, **ex**ternally is transformed into a circle, $q'(Q', R')$, that touches the circle of inversion, c, **in**ternally at a point, T', which is a diametrically opposite image of the tangency point, T.

Proof. Exercise (Note that in this theorem the circles q and q' from the Theorem 27 simply trade places. For a proof of this theorem rename the circle q into q' and conversely, rename the circle q' into q, rename the points A and T into A' and T' correspondingly and conversely, rename the points A' and T' into A and T correspondingly). □

Theorem 29. Under the inversion of a plane with respect to a circle, $c(O, r)$, a circle, $q(Q, R)$, that does not pass through the center of inversion, O, has exactly zero points common with the circle of inversion, c, and is located inside of that circle is transformed into a circle, $q'(Q', R')$, that does not pass through the center of inversion, O, has exactly zero points common with the circle of inversion, c, and is located outside of that circle (Figure 7.7):



Verbally this theorem sounds the same for inversions with both positive and negative powers. The only difference between these powers is the location of the image circle, q'.

For an inversion with positive power the image circle, q', and the circle of inversion, c, are located on the same side of the center of inversion, O, while for an inversion with negative power these two circles are located on the opposite sides of the center of inversion.

Above, in Figure 7.7, we show a sample drawing applicable to an inversion with negative power. Our readers are encouraged to explore the case of the positive power of inversion themselves.

Proof. Exercise. □

From the above theorem it immediately follows that in order to construct the image of a circle $q(Q, R)$ under the inversion of a plane with respect to a circle $c(O, r)$ with positive/negative power when the circle q does not pass through the center of inversion, O, has exactly zero points in common with the circle of inversion, c, and is located inside of that circle with Euclidean tools alone, we carry out the following steps.

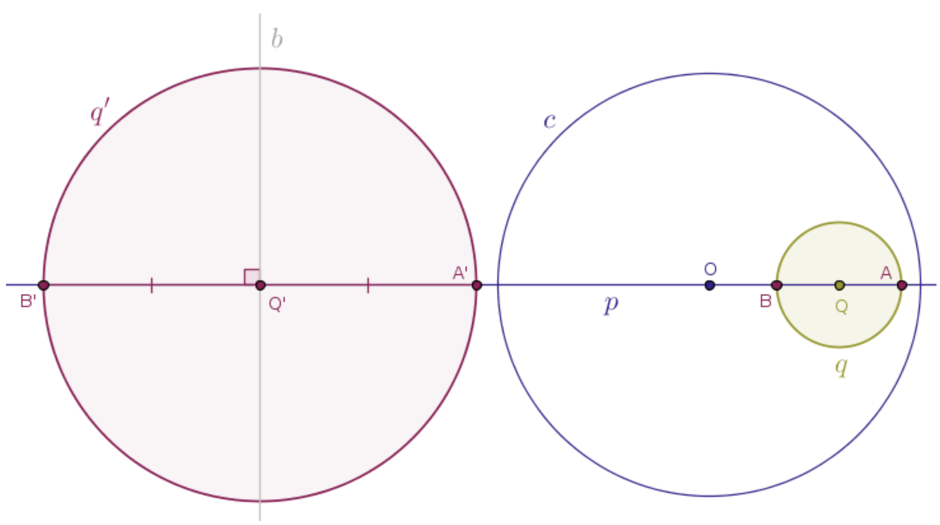
Draw a straight line, p, through the center of inversion, O, and the center, Q, of the given circle, q.

That straight line (of centers), p, will intersect the given circle, q, at two distinct and diametrically opposite points, A and B.

Reflect the above points A and B in the circle of inversion, c, with positive/negative power into its image point, A' and B'.

Bisect the line segment $A'B'$ with a straight line, b, until that straight line intersects the line of centers, p, at a point, Q', that is the center of the sought-after image circle, q'.

Draw the circle, q', centered at the point said point Q' with the radius $|Q'A'| = |Q'T'|$ (Figure 7.8)



The circle $q'(Q', |Q'A'| = |Q'T|)$ thus constructed is the image of the circle q (in the above Figure 7.8 we show a sample construction of the image circle when the power of inversion is negative).

Proof. Exercise. □

The following theorem is the converse of the Theorem 28.

Theorem 30. Under the inversion of a plane with respect to a circle, $c(O, r)$, a circle, $q(Q, R)$, that does not pass through the center of inversion, O, has exactly zero points common with the circle of inversion, c, and is located **out**side of that circle is transformed into a circle, $q'(Q', R')$, that does not pass through the center of inversion, O, has exactly zero points common with the circle of inversion, c, and is located **in**side of that circle.

Proof. Exercise (Note that in this theorem the circles q and q' from the Theorem 28 simply trade places. For a proof of this theorem rename the circle q into q' and conversely, rename the circle q' into q, rename the points A and B into A' and B' correspondingly and conversely, rename the points A' and B' into A and B correspondingly). □

As we noted several times already, observe that in general under an inversion of a plane with respect to a circle with either power, positive or negative, the center of the preimage circle is Not (!) taken into the center of the image circle.

Consequently, the next theorem establishes a relationship between the radii of two circles that are the inversive images of each other.

Theorem 31. If $q'(Q', R')$ is a circle inverse to the circle $q(Q, R)$ under the inversion of a plane with respect to a circle $c(O, r)$ and $d = |OQ|$ then it is the case that:

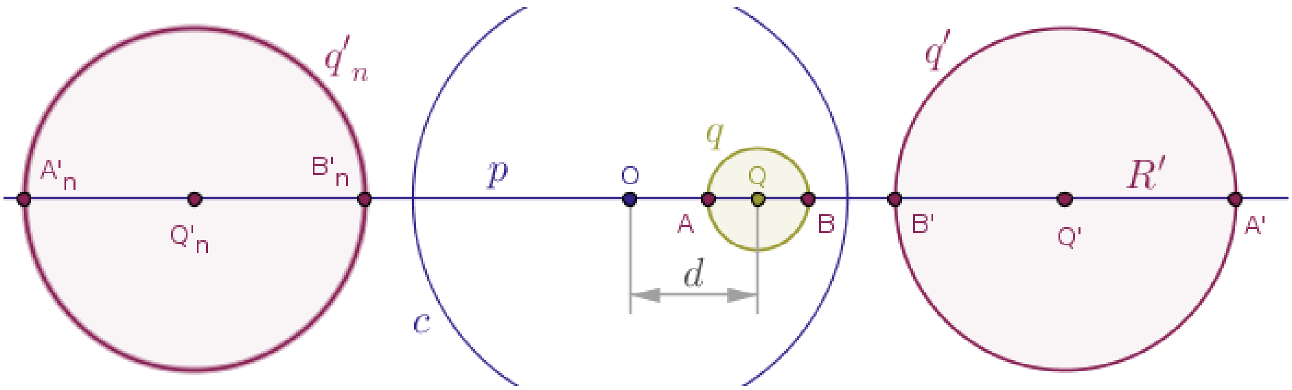
$$R' = R \cdot \frac{r^2}{|d^2 - R^2|} \tag{1}$$

Again, note that verbally this theorem sounds the same for inversions with both positive and negative powers. The only difference between these powers is the location of the image circle, q'.

For an inversion with positive power the image circle, q', and the circle of inversion, c, are located on the same side of the center of inversion, O, while for an inversion with negative power these two circles are located on the opposite sides of the center of inversion.

Below we show a complete proof applicable to an inversion with positive power and we expect that our readers will generate the relevant proof for the negative power case themselves.

Proof. Let $q'(Q', R')$ be a circle inverse to the circle $q(Q, R)$ under the inversion of a plane with respect to a circle $c(O, r)$ with positive power and let it be given that the distance between the center of inversion, the point O, and the center, the point Q, of the given circle, q, is known as $|OQ| = d$ (Figure 7.9):



From the figure above it should be evident that since $R' = |Q'A'|$, it is the case that:

$$R' = |Q'A'| = \frac{|B'A'|}{2} \tag{2}$$

because the line segment $B'A'$ is a diameter of the image circle, q'.

But that diameter, as a linear distance, can also be expressed as the following distance:

$$|B'A'| = |OA'| - |OB'|$$

and hence:

$$R' = \frac{|OA'| - |OB'|}{2}$$

By the Condition 3 of Definition 1, for the above points A and B we must have:

$$|OA'| = \frac{r^2}{|OA|}, \quad |OB'| = \frac{r^2}{|OB|}$$

Thus, for the radius, R', of the image circle, q', we have:

$$R' = \frac{r^2}{2} \cdot \left(\frac{1}{|OA|} - \frac{1}{|OB|} \right)$$

which is to say that:

$$R' = \frac{r^2}{2} \cdot \frac{|OB| - |OA|}{|OA| \cdot |OB|}$$

But it is trivially the case that $|OA| = d - R$, while $|OB| = d + R$. Hence:

$$R' = \frac{r^2}{2} \cdot \frac{d + R - d + R}{|d^2 - R^2|} = \frac{r^2}{2} \cdot \frac{2R}{|d^2 - R^2|}$$

from where our claim follows:

$$\boxed{R' = R \cdot \frac{r^2}{|d^2 - R^2|}}$$

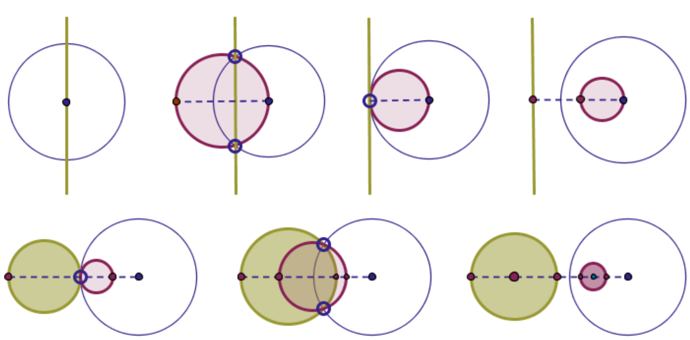
Observe that as $d \to R$, the magnitude of the length of the radius R' of the image circle, as per (**1**), grows without bound as it tends to $+\infty$:

$$R' = \lim_{d \to R} R \cdot \frac{r^2}{|d^2 - R^2|} = +\infty$$

This fact is consistent with our earlier findings because if $d \to R$, then it means that the original circle q passes through the center of inversion O, in which case its various images, as per the previously proved Theorems 15 through 19, are various **straight lines**.

Thus, we see how the following geometric idea emerges: a straight line is a circle that passes through a given point and a point at infinity. □

Before we move on to the next chapter, we show a pictorial summary of popular inversions with positive and negative powers that we captured officially in all the theorems that we have proved so far. In the diagrams below the circle of inversion is always shown in dark blue, while the relevant images are shown in olive and purple. Here are popular inversions with positive power (Figure 7.10):



and here are popular inversions with negative power (Figure 7.11):